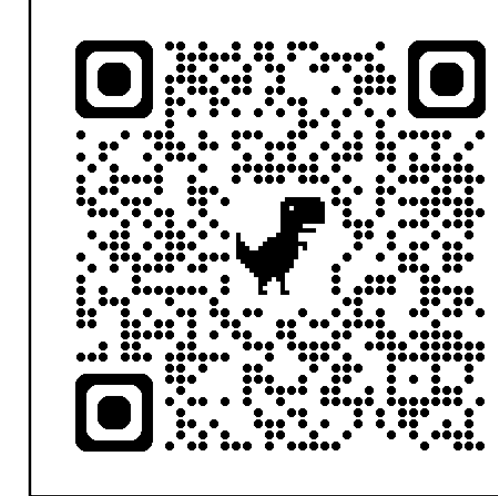


Accessing Mg-Ion Storage in V_2PS_{10} via Combined Cationic-Anionic Redox

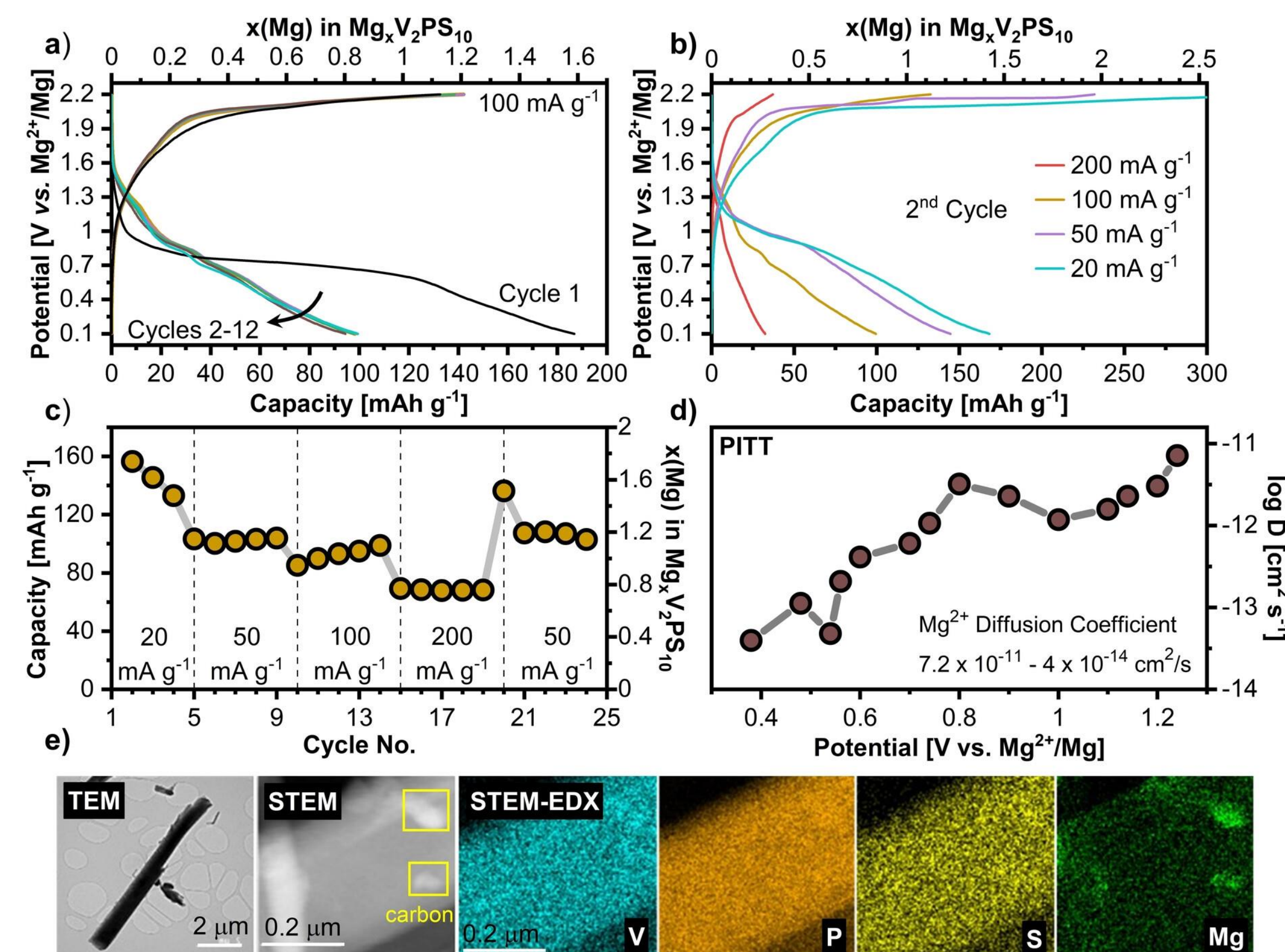
Anionic Redox • Magnesium Batteries • X-ray Diffraction • X-ray Spectroscopy • Diffusion Kinetics • Density Functional Theory

MA Wright, AV Chadwick, JB Claridge, L Hardwick, MJ Rosseinsky* & coworkers, *Angew. Chemie. Int. Ed.*, 2024, **63**, 18, e202400837

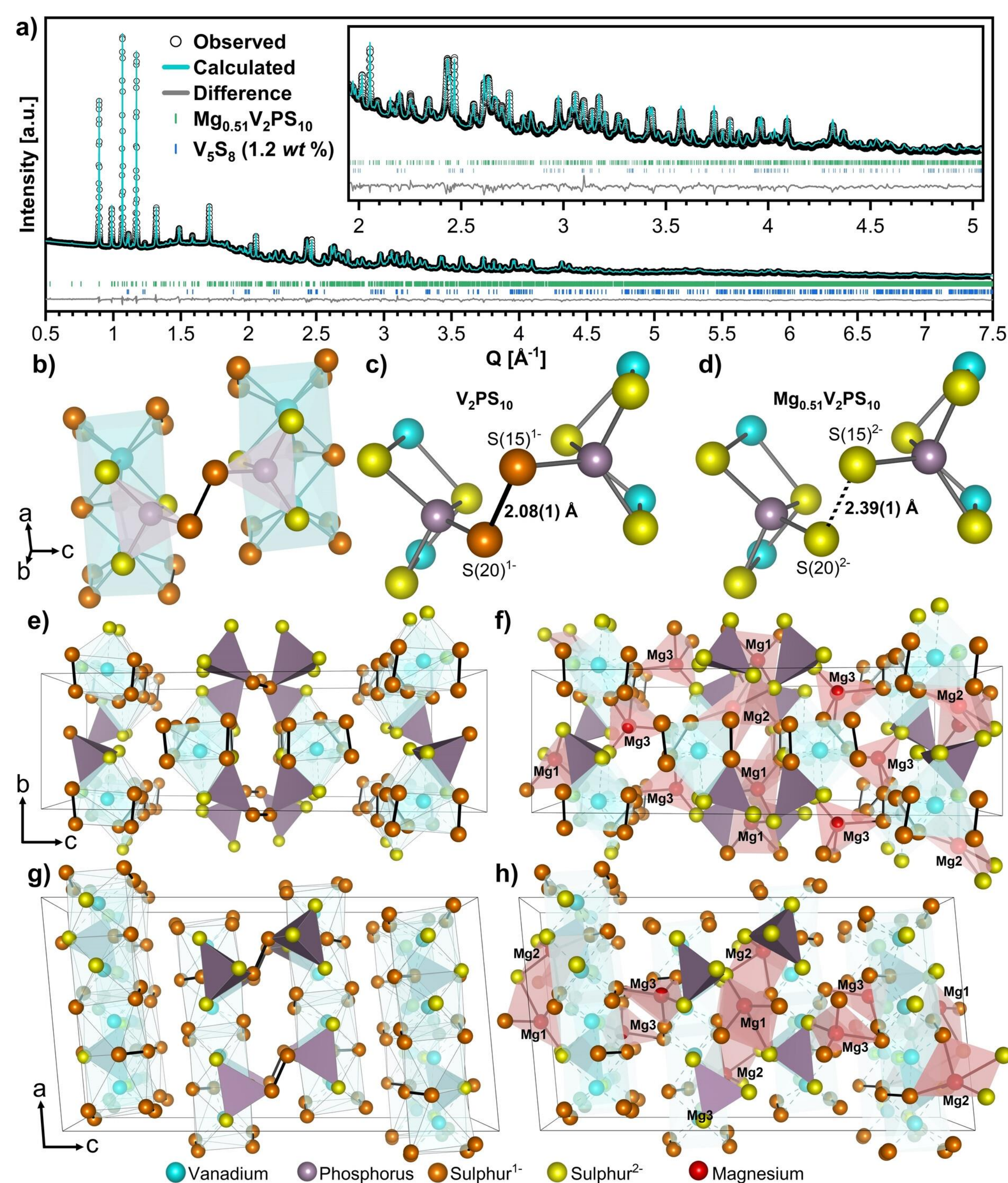
Materials Innovation Factory & Stephenson Institute for Renewable Energy, Department of Chemistry, University of Liverpool, L69 7ZD, United Kingdom



Overview: Selective, bond-specific anion redox spatially separated from transition-metal redox provides a route to multivalent-ion insertion cathodes that combine multi-electron capacity, fast early kinetics, and low structural strain.

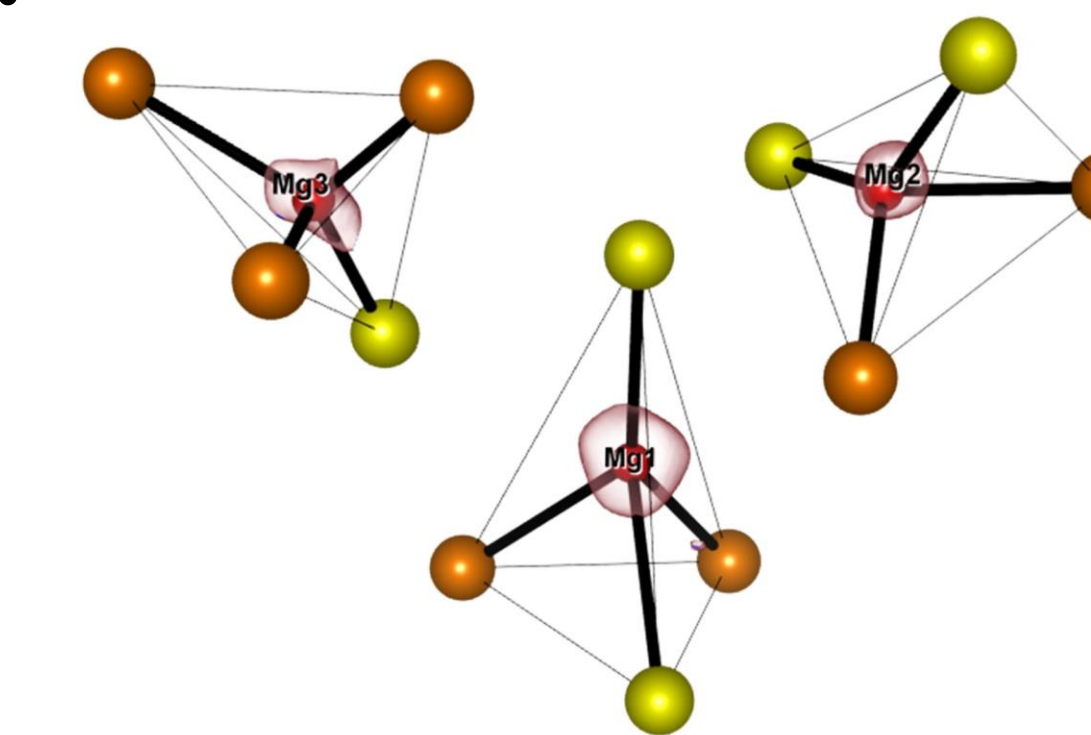


- **Three distinct Mg tetrahedral sites** identified by SXRD Rietveld refinement, Fourier difference analysis, and maximum-entropy method calculations; occupancies follow $Mg(1) > Mg(2) > Mg(3)$ with total x up to ≈ 0.51 confirmed by STEM-EDX and capacity.
- **Bond cleavage & reversibility:** $S(15)-S(20)$ distance increases from $\sim 2.08 \text{ \AA}$ to $\sim 2.40 \text{ \AA}$ on discharge and returns on recharge, showing reversible S-S cleavage and reformation.

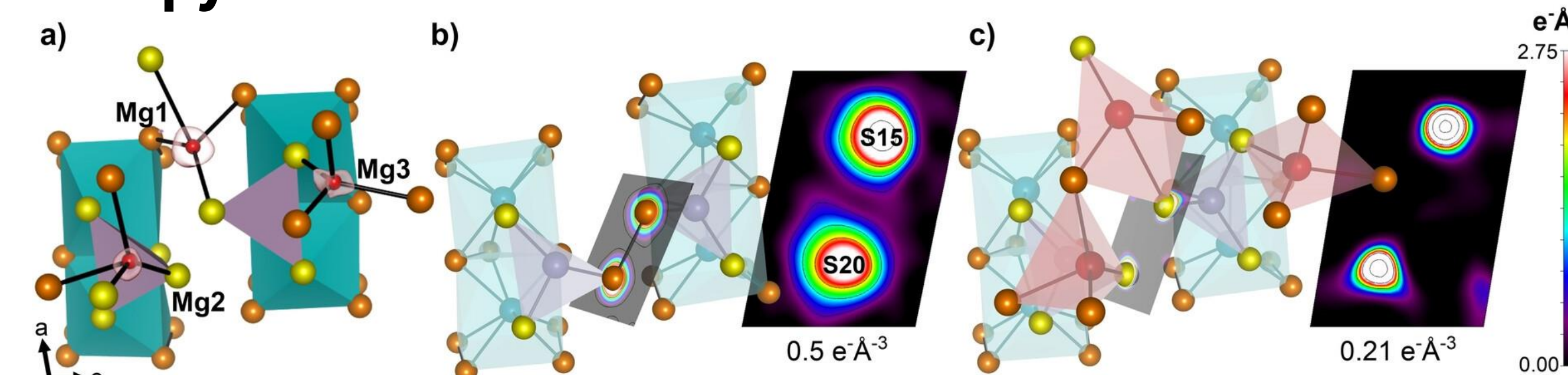


- **Step 1 – Early insertion [Mg(1) & Mg(2)]:** selective cleavage of the bridging $(S_2)^{2-}$ $S(15)-S(20)$ bond forms S^{2-} anions that create tetrahedral $Mg(1)/Mg(2)$ coordination sites; this anionic reduction is coupled to partial oxidation of $V^{4+} \rightarrow V^{5+}$, producing new S^{2-} that coordinate Mg and connect chains via S-Mg-S bonds.
- **Step 2 – Later insertion (Mg(3)):** Mg occupies a vdW-gap tetrahedral site ($Mg(3)$) that does not require additional S-S cleavage; insertion here transfers electrons back to V, partially reducing $V^{5+} \rightarrow V^{4+}$.
- **Net effect:** spatial separation of anionic and cationic redox enables multi-electron storage with reversible insertion and low structural perturbation.

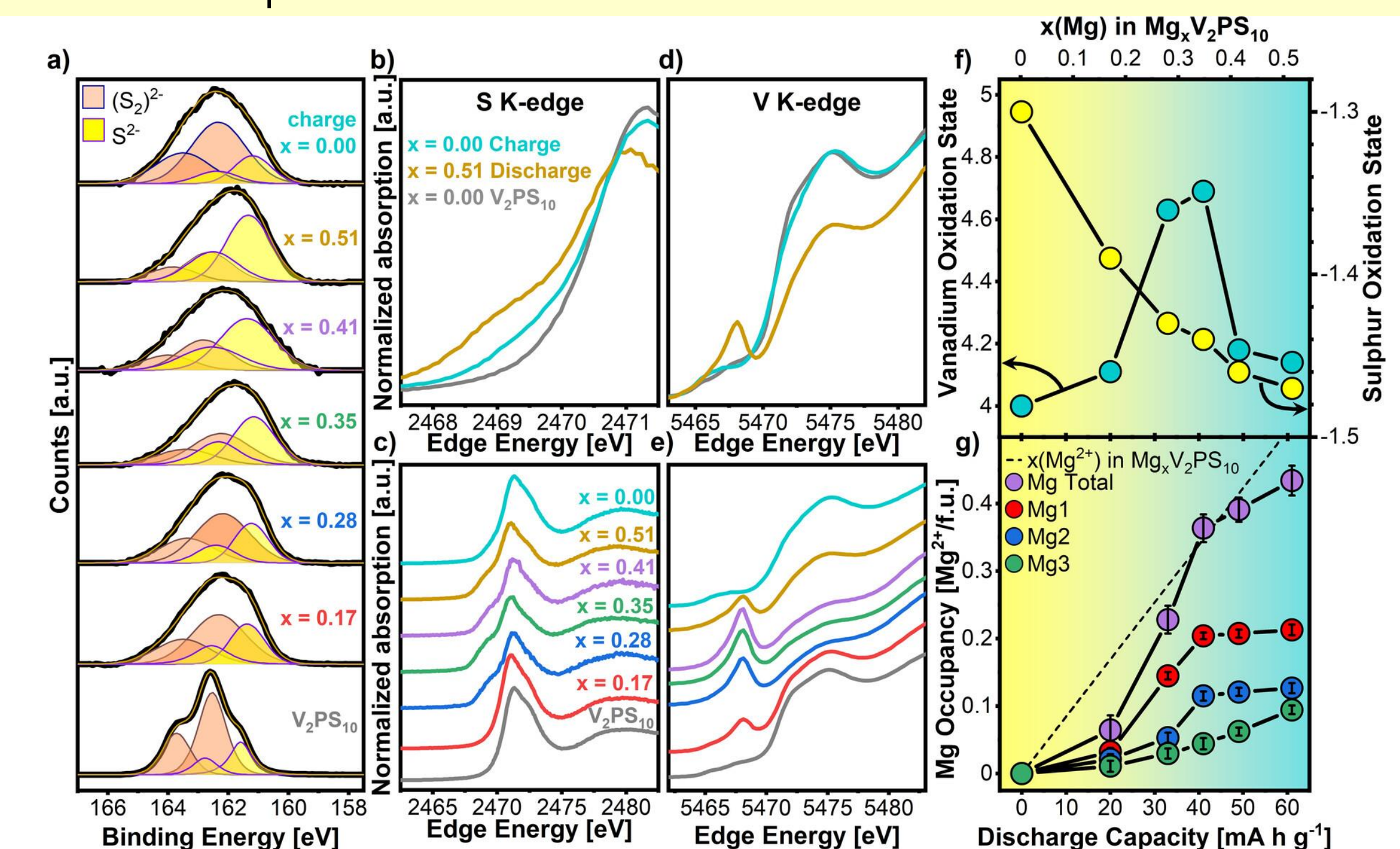
Fourier Difference Analysis:



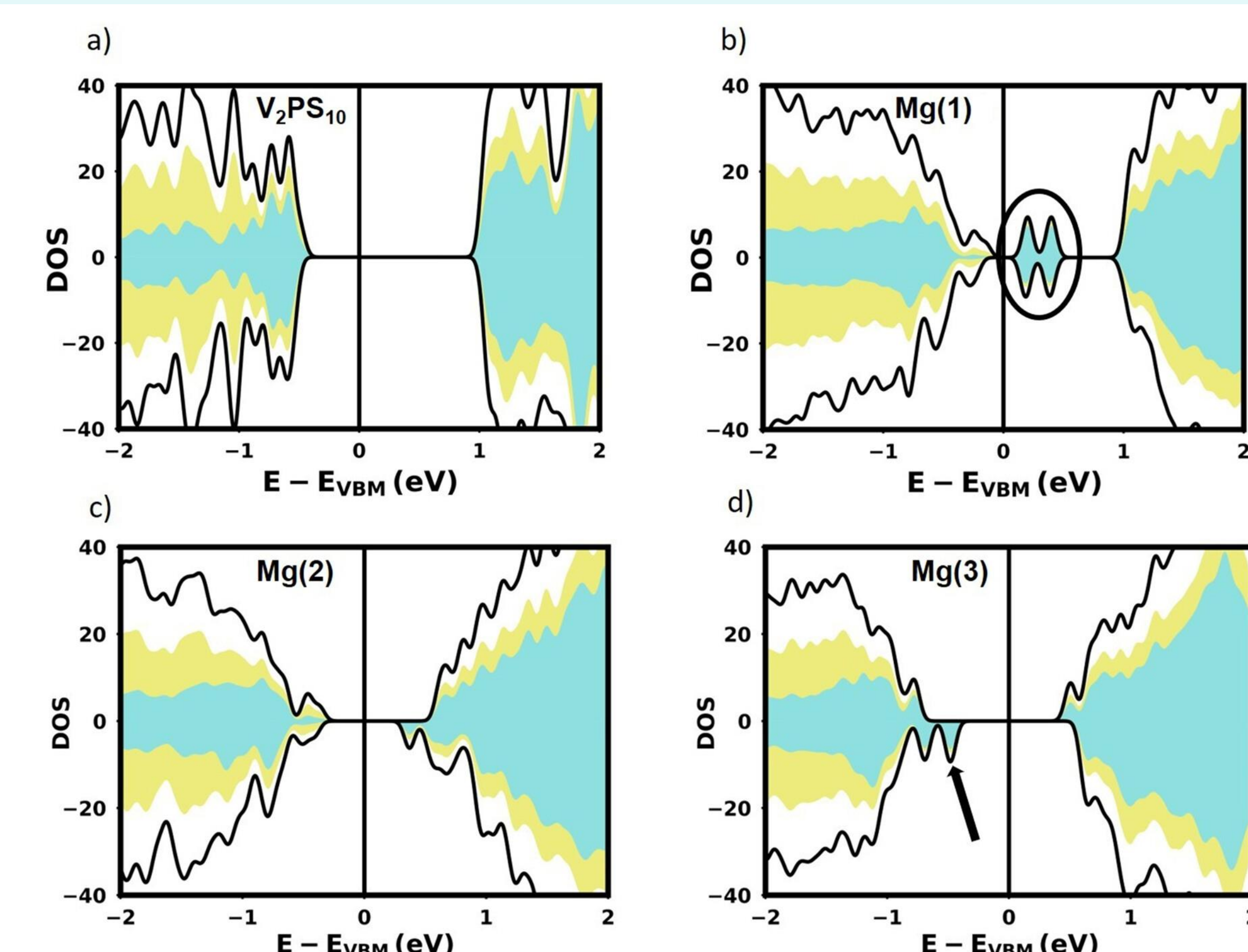
Maximum Entropy Method:



- **XPS & XANES:** S 2p and S K-edge shift consistent with $(S_2)^{2-} \rightarrow 2S^{2-}$ reduction on discharge and reversal on charge; V K-edge shows $V^{4+} \rightarrow V^{5+}$ during early insertion then partial re-reduction on deeper insertion



- **Site-dependent charge compensation:** DFT shows $Mg(1)$ occupation energetically favors breaking two bridging S-S bonds with electron transfer from V to S, creating empty V 3d states (V oxidation).
- **Mg(2) vs Mg(3):** $Mg(2)$ requires breaking only one bridging S-S (no net $V \rightarrow S$ electron transfer predicted), while $Mg(3)$ insertion reduces V and creates new occupied V d states without S-S cleavage.
- **pDOS picture:** pristine material has mixed V 3d and S 3p at the valence edge; Mg occupation redistributes states consistent with the experimental XAS trends.



Takeaway: V_2PS_{10} demonstrates reversible Mg^{2+} insertion enabled by cooperative, site-dependent anionic and cationic redox where selective, reversible cleavage of bridging S-S bonds creates Mg coordination sites while preserving V-S chain integrity, yielding fast initial diffusion, low lattice strain, and stable cycling near 100 mAh g^{-1} , offering a transferable motif for future multivalent cathode design